

Exploring Data Bias in the UCI Adult Census Income Dataset

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- 1. Introduction:** One of the most significant issues faced in today's data science field is the presence of Bias in the dataset. The use of such data in training machine learning models reinforces the existing bias, which has some serious consequences in fields such as recruitment, loan approval, and criminal sentencing. This report aims to provide a systematic analysis of the bias present in the UCI Adult Census Income dataset. This dataset is very commonly used in research and fairness analysis.
- 2. Dataset Overview:** The UCI Adult Census Income dataset was selected from the database Collected by the US in 1994 by Ronnie Kohavi and Barry Becker. Dataset has a total of 48,842 records with 15 features ranging from age to work hours to whether an individual belongs to the $\leq \$50K$ class or the $> \$50K$ class. Other features include occupation, relationship, education, race, sex, marital status and native country. The data has a considerable degree of class imbalance, whereby 76.1% of the records belong to the $\leq \$50K$ class while 23.9% belong to the greater than $> \$50K$ class (3.18:1 ratio).
- 3. Methods:** The tools for analysis were python along with pandas, numpy, Matplotlib, and seaborn libraries. Both training and test splits were loaded and merged into a single 48,842 observation database so that we have a population level dataset. We use the following techniques:
 - Descriptive statistics and visualization of income rate distributions within different demographics.
 - Missing Data Analysis to determine whether any of the missing data is correlated with any of the attributes.
 - Education controlled comparisons to test whether income gap exists beyond education level.
- 4. Results:**

1. Gender Bias: This is clearly evident from the income difference by sex. The percentage of males earning more than \$50K is 30.38, whereas the percentage of females earning more than \$50K is only 10.93, which gives a difference of 19.45%. The DIR is 0.36, which means that females gain access to high income at 36% of the rate of males. It is well below 0.80, which means that there is significant disparity. Even when considering all the education levels the gap is consistent, where any females who holds a doctorate or professional degrees earn lower income than males.

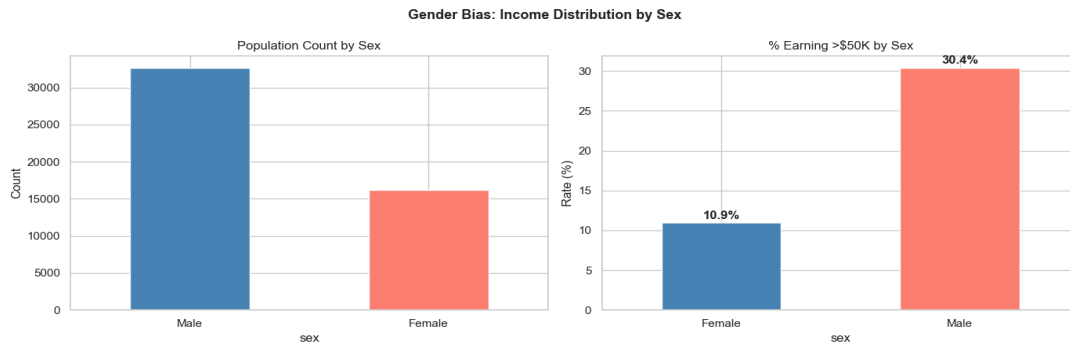


Fig: Population Count by Sex and % Earning >\$50K by Sex

2. Missing Data Bias: The dataset contains 3620 rows (7.4%) missing values, most of them found in the workclass (2799 missing), occupation (2809 missing) and native country (857 missing). The missing values in the variable “occupation” range between 7.86% females and 4.70% for males and across races ranging from 5.55% (White) to 7.64% (Other). The rows with missing values have a higher rate of low income as opposed to the other rows without missing values.



Fig: % Missing Occupation by Race and % Missing Occupation by Sex

3. Education Bias: Education came out as one of the factors with the strongest association with income in the dataset. It was evident that individuals with higher levels of education tended to earn more than USD 50,000. Around 74.9% of Professional School degree holders and 74.7% of Doctorate holders earned incomes higher than USD 50 000 while just 16.4% of High School graduates did. This trend implies that the level of education one attains is closely related to the potential for a higher income. The Chi Square test was used to prove this relationship. The test returned a Chi Square value of 4070.38 with a p value less than 0.001 suggesting In fact that a statistically significant relationship exists between education levels and income.

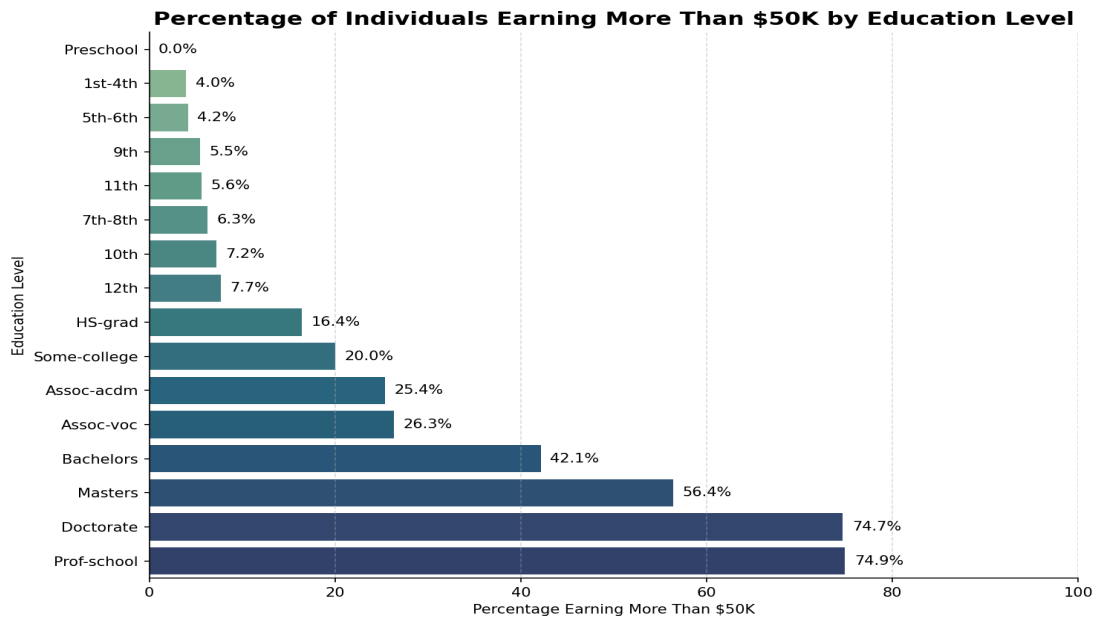


Fig: Income Distribution by Education Level and Percentage Earning More Than USD 50,000

4. Native Country Representation Bias: As the data is mostly from people in the US, we can say that more than 27,500 are from the US. The next in line after the US is Mexico, but only with 610 data records. Many other countries are represented even less, as the number is below 100. This kind of distribution of the data is highly uneven. On one hand this can lead us to problems in building a machine learning model, because the model would learn mainly from the data of one group, thereby making it impossible to generalize the model's performance across various populations. Because of this, we can see that before employing a dataset for any predictive modeling it is really essential to have a good understanding of its contents.

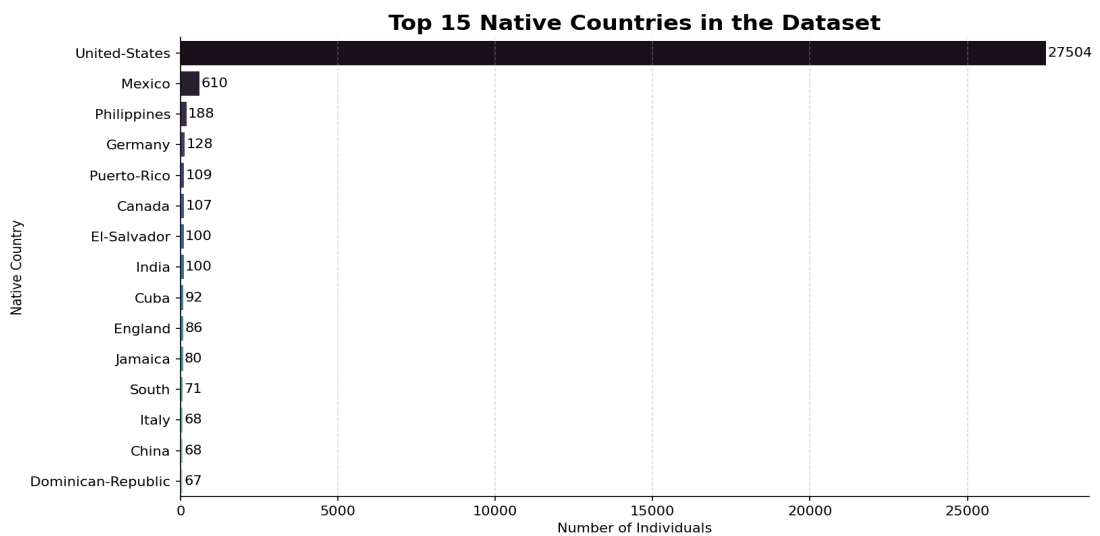


Fig: Top 15 Native Countries in the Dataset

- 5. Ethical Implications:** The findings of this study confirm that bias may already be present in datasets when a machine learning model is not even built yet. The disparities in variables of education level gender missing data, and country representation, which have been found here through this work, can cause unfairness in the prediction made by the models and because of this might be a channel for perpetuating the current social-economic inequalities. One of our contributions is to draw attention to the significance of dataset evaluation before its use in machine learning. Data scientists are expected to analyze the level of representation, figure out the possible channels of bias, and make a decision to go for suitable solutions every time. Designing fairness and transparency in the AI systems depends on the first step, that is, knowing the data's drawbacks, thereby dataset evaluation plays a very big role in the development of responsible and ethical AI.
- 6. Conclusion:** This effort investigated various aspects of bias inherent in the UCI Adult Census Income dataset using exploratory data analysis, visualization, and statistical testing. Our results proved that factors like sex, level of education, missing data, and national origin are all related to income disparity which may also affect machine learning models. In general, this work is an indication that knowing the data as much as possible is of equal importance to building the models itself. Detecting possible bias early helps data scientists choose wisely, increase fairness of models, and create more transparent, dependable, and ethically responsible AI systems.